

Rugveda Desai

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EDUCATION

Manipal Institute of Technology, Bangalore

B.Tech in Computer Science and Data Science; CGPA: 8.23

Bangalore, Karnataka

July 2024 – May 2028

EXPERIENCE

Cognifyz Technologies

Data Science Intern

Maharashtra

Summer 2025

- Conducted data preprocessing, feature engineering, EDA, and visualization on structured datasets using Python data science libraries.

Manipal Institute of Technology

Research Intern

Bangalore, Karnataka

March 2025 – Present

- Researched LLM adversarial attacks, prompt injection, and model robustness while contributing to survey publications on foundation model security and AI-driven forecasting methods.

PUBLICATIONS & RESEARCH HIGHLIGHTS

Adversarial Attacks on Foundation Models: Accepted for peer-reviewed oral presentation at the Second International Conference on Artificial Intelligence, Computation, Communication and Network Security, University of Wollongong Dubai, UAE.

Wellora: Selected under **Yukti Innovation (Government of India)** for presentation at the **IIC Regional Meet**, Bengaluru.

PROJECTS

CollabNote: Real-Time Collaborative Note Editor

June 2026

React, TypeScript, Node.js, Socket.io, PostgreSQL, Redis, Docker

- Built a real-time collaborative editor with Socket.io supporting multi-user editing, live cursors, and user presence tracking.
- Optimized performance with debounced auto-save and Redis pub/sub architecture, reducing database writes and enabling horizontal scaling
- Implemented JWT authentication, PostgreSQL data models, and Dockerized full-stack deployment on Railway.

DevPlus: AI-Powered Developer Analytics Platform

July 2026

React, TypeScript, Node.js, Express.js, PostgreSQL, Prisma, GitHub API, Gemini AI

- Built an AI-powered GitHub analytics platform that aggregates repository activity, commits, pull requests, programming languages, and contribution metrics to generate comprehensive developer profiles
- Implemented a scalable backend using Express.js, PostgreSQL, Prisma ORM, intelligent caching, and Gemini AI to deliver developer scores, personalized insights, strengths, weaknesses, and repository analytics through REST APIs

PrepPal: AI-Powered Mock Interview Platform

July 2026

Next.js, TypeScript, FastAPI, PostgreSQL, OpenAI API, Docker

- Engineered a full-stack AI interview platform with role-aware question generation, enabling personalized mock interviews across 8 professional domains using GPT-4o.
- Built real-time streaming AI evaluation with Server-Sent Events (SSE), delivering structured interview feedback, scoring rubrics, strengths, gaps, and improved responses through an interactive chat-like experience.
- Architected secure JWT authentication, PostgreSQL data persistence, browser-native voice transcription, interview performance analytics, and containerized deployment using Docker.

Wellora: Biomedical Mental Health Platform

April 2025 – July 2025

Research Integration, Python, Dart, ML, System Design

- Developed a biomedical mental health platform integrating ML-based features with a modular full-stack application architecture.
- Translated clinical research requirements into scalable technical specifications, workflows, and system components while collaborating with stakeholders.

TECHNICAL SKILLS

Programming Languages: JavaScript, TypeScript, Python, Java, SQL, HTML/CSS

Frontend: React, Next.js, Tailwind CSS, Vite, Context API, React Hooks, Tiptap Editor

Backend: Node.js, Express.js, FastAPI, REST APIs, JWT Authentication, Middleware Design, Socket.IO

Databases & ORM: PostgreSQL, Prisma ORM, MongoDB

Cloud & DevOps: Docker, Docker Compose, Git, GitHub, Postman, Vercel, Render, Cloudinary

AI/LLM: OpenAI API Integration, Prompt Engineering, RAG Fundamentals, Vector Embeddings

Data Science & Visualization: Pandas, NumPy, Matplotlib, Seaborn, OpenCV, Streamlit

LEADERSHIP & EXTRACURRICULARS

President, CodeX Tech Club: Led research-oriented technical initiatives and mentored peers on SWE project development

Organizing Committee Tech Solstice : a campus-wide technical event with 2000+ participants